2017 HW1

MOS Theory, Dual Thresholds, and CMOS Logic

EE3032 / Introduction to VLSI Design / Jin-Fu Li

The original two-page prompt and supplied scanned answers were read first. The results below are independently derived; the source ledger identifies every reference. PMOS bodies connect to VDD and NMOS bodies to VSS. True external literals suffice for the static-CMOS networks.

1. Two identical series devices per inverter branch

(a) The rail-side devices M_1 and M_4 cannot carry nonzero DC current in saturation. For the nMOS stack let the internal node be u and $U = V_{\text{in}} - V_{tn} > 0$. Saturation of the lower M_1 would require $u \geq U$. But then the upper device has $V_{GS,2} = V_{\text{in}} - u \leq V_{tn}$ and is cut off. Thus the assumed positive stack current is impossible. The pMOS argument is identical with voltages measured downward from V_{DD}: saturation of rail-side M_4 would cut off M_3. At cutoff or the zero-current boundary, the usual region inequalities may coincide; the statement concerns conducting operation. **(b) Both equivalent aspect ratios are one half of the original:**

$$\boxed{(W/L)_{N_1} = (W/L)_{N_2} = \tfrac{1}{2}(W/L).}$$

Here N_1 is nMOS and N_2 is pMOS. This equality uses the ideal long-channel square-law model, no body effect, and no channel-length modulation; it does not assert exact BSIM short-channel equivalence. Let $k_n = \mu_n C_{\text{ox}}(W/L)$. If the upper nMOS is saturated,

$$k_n(Uu - u^2/2) = \frac{k_n}{2}(U-u)^2, \quad u = U\left(1 - \frac{1}{\sqrt{2}}\right), \quad I = \frac{k_n U^2}{4}.$$

A single saturated nMOS with $k_{\text{eq}} = k_n/2$ has the same current. If both devices are linear and $v = V_{\text{out}} < U$, adding their equal-current equations gives

$$\frac{2I}{k_n} = Uv - \frac{v^2}{2}, \qquad I = \frac{k_n}{2}\left(Uv - \frac{v^2}{2}\right).$$

Thus the half-aspect-ratio device also matches the linear part of the branch characteristic; the pMOS derivation is symmetric. Halve W at fixed L, or double L in this ideal model. **Reference comparison.** The supplied answer identifies M_1, M_4 and the factor $1/2$ correctly. The present proof also covers the linear region. `verify_logic.py` checks 5,050 normalized operating points; the maximum current-equation error is below 10^{-12}.

2. High-V_t selection on the NAND branch

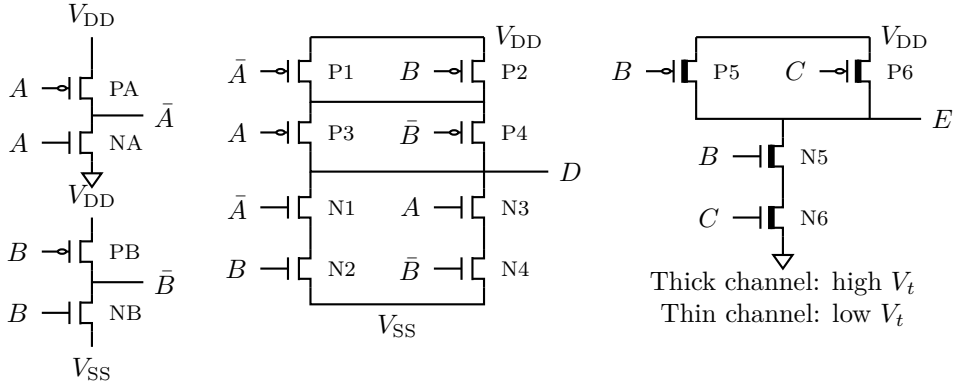
From the original gate symbols,

$$D = \bar{A} \oplus B = AB + \bar{A}\bar{B}, \qquad E = \overline{BC}.$$

With the given one-unit low-V_t gate model, the path $A \to \textit{inverter} \to XOR \to D$ is critical at two units. The direct $B \to XOR \to D$ and the two NAND paths each take one unit. Keep the inverter and XOR realization low-V_t; replace the *entire* NAND driving E by high-V_t devices.

Path	Low-V_t delay	New delay	Slack before change
$A \to D$	2	2	0
$B \to D$	1	1	1
$B, C \to E$	1	$d_{\mathrm{HVT,NAND}}$	1

Necessary timing condition: $d_{\mathrm{HVT,NAND}} \leq 2$ units. The prompt supplies no high-V_t delay or threshold values, so a numerical leakage reduction or unconditional timing guarantee cannot be inferred. Size the high-V_t NAND if necessary to satisfy this condition. Changing transistors inside the shared critical XOR without a library timing model is not justified by the direct-B slack alone. Below is a fully complementary static-CMOS realization. Complement-driver outputs are labeled rather than routed across the schematic; connect each identically named gate to its driver. All pMOS bodies connect to VDD and all nMOS bodies to VSS. The B complement generator belongs to the XOR-cell implementation, within the given abstract cell-delay model.



Reference comparison. The supplied scan correctly makes the noncritical NAND high-V_t and also suggests optional high-V_t devices inside the XOR. We retain the NAND change. The optional internal changes need a characterized cell timing model to guarantee the critical A path; the given direct-B slack alone is insufficient. All eight Boolean combinations are checked independently. CIC018 has no supplied characterized dual-V_t variants; no process leakage value is invented.

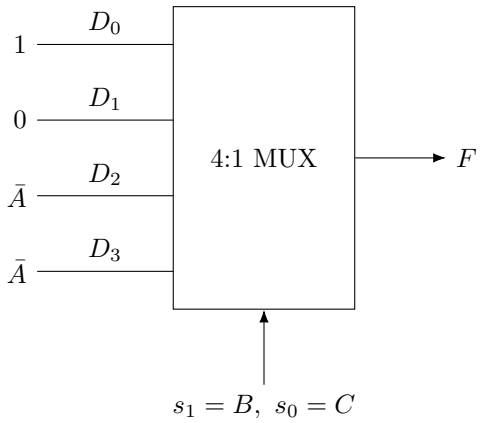
3. One 4:1 multiplexer

$$F = \bar{A}(B + \bar{C}) + A\bar{B}\bar{C}.$$

Choose the select pair $(s_1, s_0) = (B, C)$. Its Shannon cofactors are:

B	C	Data input
0	0	$D_0 = 1$
0	1	$D_1 = 0$
1	0	$D_2 = \bar{A}$
1	1	$D_3 = \bar{A}$

Thus one CMOS inverter generates $\bar{A}$, which drives both D_2, D_3. The 4:1 mux is selected in binary order 00,01,10,11:



A	B	C	F
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	0

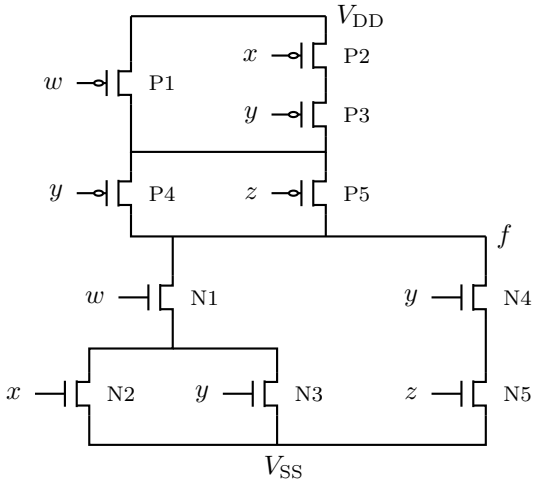
Reference comparison. The supplied scan gives three valid select-pair choices, including this BC choice. We retain it because it uses only one complement and two repeated data inputs. All eight cases pass the independent cofactor check.

4(a). Complemented AOI function

Read the overbar in the prompt carefully: it spans the entire sum.

$$\boxed{f(w,x,y,z) = \overline{w(x+y)+yz}.}$$

The pull-down network (PDN) implements $G = w(x+y)+yz$: n_w in series with $(n_x \parallel n_y)$, in parallel with (n_y series n_z). The pull-up network (PUN) is the dual: $(p_w \parallel (p_x \text{ series } p_y))$ in series with $(p_y \parallel p_z)$.



There are five nMOS and five pMOS devices: **10 transistors total**. Two devices are separately controlled by y in each network; they are distinct transistors.

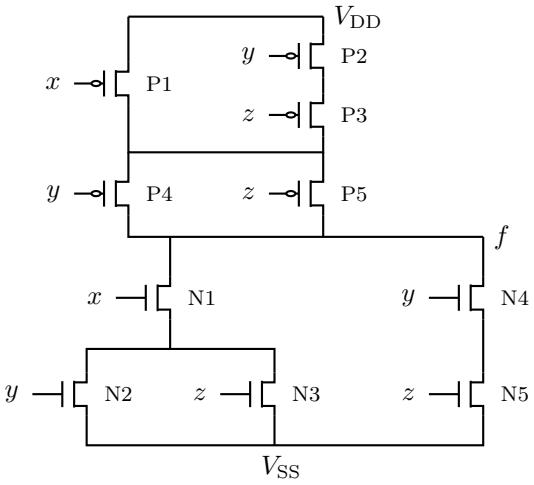
$$PUN = [\bar{w} + \bar{x}\bar{y}](\bar{y} + \bar{z}) = \overline{w(x+y)+yz} = f.$$

Every input has exactly one conducting rail network in ideal switch logic, so there is no static VDD-to-ground path. No complemented external input is needed. **Reference comparison.** The cleaned scan uses the same correct 10-transistor topology. The photographed answer reads the polarity differently and adds complemented inputs; we follow the original full overbar. This solution retains it, clarifies the overbar, and checks all 16 input combinations.

4(b). Complement of a three-input majority

$$f = \bar{x}\bar{y} + \bar{x}\bar{z} + \bar{y}\bar{z} = \overline{xy + xz + yz} = \boxed{\overline{x(y+z) + yz}}.$$

Use the same network structure as 4(a), now with the first branch n_x in series with $(n_y \parallel n_z)$ and the second branch n_y in series with n_z.



Gate labels in this diagram. P4/N4 are controlled by y and P5/N5 by z; repeated labels denote independent devices. The dual PUN is $(p_x \parallel (p_y \text{ series } p_z))$ in series with $(p_y \parallel p_z)$. It uses true literals only and needs **10 transistors**.

x	y	z	f
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	0

Reference comparison. The supplied answer implements the equivalent PDN $(x+y)(x+z)(y+z) = xy + xz + yz$, using six nMOS and six pMOS. Factoring the majority expression saves two transistors. All eight combinations pass the independent PDN/PUN complement check.